



Igor Slazyk

ML Engineer | Data Engineer

Portfolio

islazykv.github.io

LinkedIn

linkedin.com/in/igor-slazyk

GitHub

github.com/islazykv

Personal Information

Nationality: Polish
Date of Birth: 03.11.1992
Address: Sandnes, Norway
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Languages

English: C1
Norwegian: A2
Italian: A2
Polish: Native

Interests

Popular Science Literature
IT / Technology
Gaming
Cycling
Hiking

Summary

ML Engineer with 8 years of experience in data science, including 4 years focused on machine learning. Experienced in building scalable data pipelines and ML systems on large and complex datasets. Driven to work at the intersection of data engineering and ML, designing pipeline infrastructure with a direct understanding of what models actually need from data.

Education

- **Ph.D. in Particle Physics** 10/2017 – 07/2021
University of Ferrara Ferrara, Italy
Institute of Nuclear Physics Krakow, Poland
- **M.Sc. in Applied Physics** 02/2016 – 05/2017
Krakow University of Technology Krakow, Poland
- **B.Sc. in Applied Physics** 09/2012 – 02/2016
Krakow University of Technology Krakow, Poland
- **B.Sc. in Civil Engineering** 09/2011 – 02/2015
Krakow University of Technology Krakow, Poland

Experience

- **Postdoctoral Researcher in ML** 01/2022 – 05/2026
Høgskulen på Vestlandet Bergen, Norway
 - Built end-to-end, multi-stage data pipelines processing datasets of 10M+ events, from raw ingestion to model-ready outputs
 - Designed scalable ETL/ELT workflows for data transformation and feature engineering
 - Implemented data validation and quality checks across pipeline stages
 - Tracked data versions, experiments and model artifacts using MLflow and W&B for full pipeline observability
 - Maintained well-tested, maintainable code with automated testing, refactoring and CI/CD via GitHub Actions
 - Communicated analytical results to cross-functional teams of engineers, analysts and domain experts
- **Affiliated Researcher** 01/2022 – 12/2025
ATLAS / CERN Geneva, Switzerland
 - Developed data transformation software within a distributed processing framework (Athena) operating at petabyte scale
 - Built and maintained modular data processing components integrated into high-throughput workflows
 - Optimized pipeline performance and refactored legacy code in a 4M-line collaborative codebase
 - Validated output data quality and interpreted results to confirm pipeline correctness
 - Collaborated with developers, analysts and domain experts on shared data tooling and infrastructure
- **Affiliated Researcher** 12/2017 – 03/2021
LHCb / CERN Geneva, Switzerland
 - Built an automated data quality monitoring system (C++/Python/LabVIEW) for continuous QA of measurement outputs
 - Developed data processing scripts and tooling for structured measurement pipelines
 - Analyzed QA output data to detect anomalies and failure patterns across large datasets
 - Explored ML-based anomaly detection for automated performance classification
- **R&D Physics Intern** 04/2016 – 09/2016
Camlin Technologies Zurich, Switzerland
 - Built data collection and processing workflows for semiconductor measurement data
 - Analyzed experimental measurement data to extract and compare material properties
 - Collaborated with software developers and physicists on shared analysis workflows

Selected Projects	
repository notebook showcase publication	Tau Supersymmetry Search Boosted Decision Trees Neural Networks - Developed a supervised ML pipeline for 6-class classification, tracked with MLflow - Trained and optimized classifiers with hyperparameter tuning via Optuna (0.97 AUC) - Implemented ML-based region construction and statistical hypothesis testing
	Tau Anomaly Detection Autoencoder Variational Autoencoder - Developed an unsupervised ML pipeline for anomaly detection, tracked with W&B - Trained and optimized models with Ray Tune (1.46× significance gain) - Served reconstruction-error anomaly scores via a REST API (FastAPI, Docker)
repository notebook showcase	Tau Energy Scale Mixture Density Network Neural Networks - Developed a probabilistic regression model with uncertainty estimation - Trained with Huber loss pretraining and NLL fine-tuning for density estimation - Evaluated model performance with energy response and resolution histograms
	Sphalerons and Black Holes Convolutional Neural Network Boosted Decision Trees - Optimized high-capacity models to classify rare, low-frequency events - Built a pipeline encoding raw detector signals as multi-channel images for CNN input - Compared high-level features with raw pixel data to maximize signal sensitivity
publication	Di-Tau Reconstruction Software Development Data Science - Engineered a C++ analysis framework for large-scale physics reconstruction studies - Benchmarked multiple algorithm configurations to optimize reconstruction efficiency - Evaluated input variations to establish performance baselines
	Elementary Cells Software Development Data Science - Developed automated C++/Python/LabVIEW software for hardware quality assurance - Analyzed array performance via noise, dark count and threshold measurements - Explored ML-based classification for detector signal identification
internal project note	
repository publication	
Technical Skills	
	Programming Languages & Tools: - Python, C++, SQL (PostgreSQL), Bash, Linux - Git, Jupyter, pytest, uv, conda
	Data Processing & Visualization: - NumPy, pandas, SciPy - Matplotlib, seaborn, Plotly
	Machine Learning & Deep Learning: - scikit-learn, XGBoost - PyTorch, PyTorch Lightning
	Tuning & Explainability: Optuna, Ray Tune, SHAP
	Data & ML Infrastructure: - Docker, GitHub Actions, FastAPI, DVC - MLflow, W&B, Hydra
	Currently Learning: PySpark, Databricks, dbt, Azure